Robotics and AI

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WHAT'S COVERED

In this lesson, you will learn about changes to the workplace and the ethical problems they present to businesses. Specifically, this lesson will cover:

1. Robotics and Automation in the Workplace

In the 21st Century workers will find their jobs aided or even replaced by automation and artificial intelligence.

While human-controlled machinery is frequently called **robotics**, the technical definition refers to machines that work autonomously (without a human operator). Advances in the field of robotics—a combination of computer science, mechanical and electronics engineering, and science—have meant that machines or related forms of automation now do the work of humans in a wide variety of settings, such as medicine, where robots perform surgeries previously done by the surgeon's hand. Even automatic teller machines and vending machines may be called "robots," in that they replace humans in simple transactions.

Robots have made it easier and cheaper for employers to get work done. The downside, however, is that some reasonably well-paying jobs that provided middle-class employment for humans have become the province of machines.

A McKinsey Global Institute study of 800 occupations in nearly 50 countries showed that more than 800 million jobs, or 20% of the global workforce, could be lost to robotics by the year 2030. The effects could be even more pronounced in wealthy industrialized nations, such as the United States and Germany, where researchers expect that up to one-third of the workforce will be affected. By 2030, the report estimates that 39 million to 73 million jobs may be eliminated in the United States. Given that the level of employment in the United States in 2023 is approximately 165 million workers, this potential loss of jobs represents roughly one-quarter to one-half of total current employment.

The big question, then, is what will happen to all these displaced workers. The McKinsey report estimates that about 20 million of them will be able to transfer easily to other industries for employment. But this still leaves between 20 million and more than 50 million displaced workers who will need new employment. Occupational retraining is likely to be a path taken by some, but older workers, as well as geographically immobile workers, are unlikely to opt for such training and may endure job loss for protracted periods.

IN CONTEXT

In 2020, presidential candidate Andrew Yang expressed concerns about the potential impact of automation and robotics on the job market. Yang believes that the rise of intelligent machines and automation could lead to widespread unemployment, especially among low-skilled workers, and a growing wealth gap. He argues that technological advancement is inevitable, but it needs to be accompanied by policies that ensure that all citizens benefit from it. To address these concerns, Yang has proposed a Universal Basic Income (UBI) as a way to provide financial security to all citizens and ensure that they can meet their basic needs even if their jobs are replaced by robots. Overall, Yang's concerns about robots highlight the need for careful consideration of the social and economic implications of technological progress.

In developing countries, the report predicts that the number of jobs requiring less education will shrink. Furthermore, robotics will have less impact in poorer countries because these nations' workers are already paid so little that employers will save less on labor costs by automating. According to the report, for example, by the same date of 2030, India is expected to lose only about 9% of its jobs to emerging technology.

Which occupations will be most heavily affected? Not surprisingly, the McKinsey report concludes that machine operators, factory workers, and food workers will be hit hardest, because robots can do their jobs more precisely and efficiently. "It's cheaper to buy a $35,000 robotic arm than it is to hire an employee who's inefficiently making $15 an hour bagging french fries," said a former McDonald's CEO in another article about the consequences of robots in the labor market. He estimated that automation has already cut the number of people working in a McDonald's by half since the 1960s and that this trend will continue. Other hard-hit jobs will include mortgage brokers, paralegals, accountants, some office staff, cashiers, toll booth operators, and car and truck drivers. The Bureau of Labor Statistics (BLS) estimates that 80,000 fast-food jobs will disappear by 2024. As growing numbers of retail stores like Walmart, CVS, and McDonald's provide automated self-checkout options, it has been estimated that 7.5 million retail jobs are at risk over the course of the next decade. Furthermore, it has been estimated that as self-driving cars and trucks replace automobile and truck drivers, 5 million jobs will be lost in the coming decade.

THINK ABOUT IT

Do employers have the responsibility to help retrain workers, particularly ones with jobs that have become obsolete?

Jobs requiring human interaction are typically at low risk for being replaced by automation. These include nurses and most physicians, lawyers, teachers, and bartenders, as well as social workers (estimated by the BLS to grow by 19% by 2024), hairstylists and cosmetologists, youth sports coaches, and songwriters. McKinsey also anticipates that specialized lower-wage jobs like gardening, plumbing, and care work will be less affected by automation.

The challenge to the economy, then, will be how to address the prospect of substantial job loss; about 20 million to 50 million people will not be able to easily find new jobs. The McKinsey report notes that new technology, as in the past, will generate new types of jobs. But this is unlikely to help more than a small fraction of those confronting unemployment. So, the United States will likely face some combination of rapidly rising unemployment, an urgent need to retrain 20 million or more workers, and recourse to policies whereby the government serves as an employer of last resort.

IN CONTEXT

Japan has long maintained its position as the world's top exporter of robots, selling nearly 50% of the global market share in terms of both units and dollar value. At first, Japan's robots were found mainly in factories making automobiles and electronic equipment, performing simple jobs such as assembling parts. Now Japan is poised to take the lead by putting robots in diverse areas including aeronautics, medicine, disaster mitigation, and search and rescue, performing jobs that humans either cannot do or cannot do safely. Leading universities such as the University of Tokyo offer advanced programs to teach students not only how to create robots but also how to understand the way robot technology is transforming Japanese society. Universities, research institutions, corporations, and government entities are collaborating to implement the country's next generation of advanced artificial intelligence robot technology, because Japan truly sees the rise of robotics as the "Fourth Industrial Revolution."

In the laboratory at the University of Tokyo School of Engineering, advances are also being made in technology that mimics the capabilities of the human eye. One application allows scientists a clear field of vision in extreme weather conditions that are otherwise difficult or impossible for humans to study.

Japanese researchers are also developing a surgical robotic system with a three-dimensional endoscope to conduct high-risk surgery in remote mountainous regions with no specialized doctors. This system is in use in operating rooms in the United States as well, but Japan is taking it a step further by using it in *teletherapy*, where the patient is hundreds of miles away from the doctor actually performing the surgery. In Japan's manufacturing culture, robots are viewed not as threats but as solutions to many of the nation's most critical problems. Indeed, with Japan's below-replacement fertility since the mid-1970s, Japan's workforce has been aging quite rapidly; in fact, the Japanese population is shrinking. Clearly, robots are potentially quite important as a means to offset prospective adverse consequences of a diminishing labor force.

THINK ABOUT IT

How might the use of robots add to the increasing inequality in the U.S. economy?

TERM TO KNOW

Robotics

The technology of machines that can work autonomously (without a human operator) to replicate the tasks of humans.

2. Artificial Intelligence

The branch of science that uses computer algorithms to replicate human intelligent behavior by machines with minimal human intervention is called **artificial intelligence (AI)**. Although AI software is sometimes called "bots," these are usually accessed by humans using computers, and only respond to human prompts, rather than carrying out tasks autonomously. Professions in which the implementation of AI might have particular impact are banking, financial advising, and the sales of securities and managing of stock portfolios. Technical support and technical writing are other professions at risk, as advanced AI language models can write expository text with some competence. Even news articles have begun to be written by AI text generators.

TRY IT

One paragraph in this tutorial was written by Chat GTP, an AI language model.

Can you spot the AI-written paragraph? +

The In Context paragraph about Andrew Yang and robotics was written by an AI and is presented here without any editorial changes. Do you see any difference in mechanics or style from the paragraphs written by humans?

According to global consulting firm Accenture, AI is "a collection of advanced technologies that allows machines to sense, comprehend, act and learn." Accenture contends that AI will be the next great advance in the workplace: "It is set to transform business in ways we have not seen since the Industrial Revolution, fundamentally reinventing how businesses run, compete, and thrive. When implemented holistically, these technologies help improve productivity and lower costs, unlocking more creative jobs and creating new growth opportunities." Accenture looked at 12 of the world's most developed countries, which account for more than half of world economic output, to assess the impact of AI in 16 specific industries. According to its report, AI has the potential to significantly increase corporate profitability, double rates of economic growth by 2035, increase labor productivity by as much as 40%, and boost gross value added by $14 trillion by 2035, based on an almost 40% increase in rates of return. But the Accenture report reflects the opportunities for business leaders and shareholders, without much consideration for any other stakeholders. Will AI increase profits at the expense of millions of office workers?

TERM TO KNOW

Artificial intelligence (AI)

The branch of computer science that replicates human intelligence with minimal human interaction.

3. Ethical Challenges of Automation and AI

A report by KPMG, another global consulting and accounting firm, indicates that almost 50% of the activities people perform in the workplace today could be automated, most often by using AI and automation technology that already exists. The ethical question facing the business community, and all of us on a broader level, is about the type of society in which we all want to live and the role automation will play in it. The answer is not simply about efficiency; a company should consider many variables as it moves toward increased automation: What are the company's core values? What will relationships be like with human stakeholders? How will the transition be implemented, and what will happen to workers displaced by AI or automation?

For example, as AI programs become better able to interact with humans, especially online, should a company be required to inform its customers if and when they are dealing with any form of AI and not a person? This issue is further muddied when a human employee largely is tapping AI to serve customers or clients. Should this combination of human and AI assistance be made clear? If a company retains its staff of technical writers, are they allowed to use AI tools to help generate content they used to write from scratch? While educators are likely to frown on the use of AI by students, what about using AI to detect AI-generated content, or even to grade papers? If teachers are allowed to use AI to write and grade lessons, is it fair that students cannot use the same technology?

As AI evolves, and becomes more and more "human," these questions will become thornier. Although traditional business ethics can provide us with a starting place to answer such questions, we will also need a philosophical approach because we are venturing into uncharted territory.

THINK ABOUT IT

> The Turing Test, or Imitation Game, is a test of a machine's ability to replicate human behavior based on a human's ability to distinguish between human and machine interactions. If people cannot tell when they are communicating with an AI program and not a human being, has the AI reached a form of personhood?

Another issue in AI and all forms of automation is liability, or legal responsibility for their actions. What used to seem like the realm of science fiction is now a pertinent question: What happens if a robot harms someone? According to Reuters News, "lawmakers in Europe have agreed on the need for [European]-wide legislation that would regulate robots and their use, including an ethical framework for their development and deployment, as well as the establishment of liability for the actions of robots, including self-driving cars."

There are also ethical concerns regarding AI's use of prior text and images that inform their models. A language model AI writer is fed a number of texts and is largely regurgitating those texts when it provides a response. AI art generators work in a similar fashion. This means that the answer it generates may include copied or slightly altered work belonging to a human creator. For companies using AI to write instruction manuals or other content, they have to be careful about inadvertently violating copyright laws.

> ↪ EXAMPLE Bestselling author Christopher Paolini's 2023 book *Fractal Noise* had cover art generated by AI, closely based on prior covers designed by human artists. What credit or compensation do those original artists deserve for the new cover?

Yet another issue is privacy, as AI can generate realistic images of people doing things they never did, such as nude (even pornographic) images of celebrities, which can be used to embarrass or blackmail them. AI imagery could even be used to fake evidence of a crime (or, for that matter, to fake an alibi). The legal and ethical questions in assigning liability for decisions made by robots and AI are not only fascinating to debate but also an important legal matter society must resolve now so the development and availability of technology is done with ethics in mind. The answers will one day directly affect the day-to-day lives of billions of people.

SUMMARY

In this lesson, you learned about how the increasing adoption of **robotics**, **automation**, and **artificial intelligence (AI)** in the workplace is leading to significant changes in how businesses operate. While these technologies can increase efficiency and productivity, they also present ethical problems for businesses to navigate. The displacement of human workers and the potential for bias in AI systems can lead to social and economic inequality. Additionally, businesses must consider the **ethical implications of using AI** such as informing people when they are interacting with an AI or AI-assisted human. Addressing these ethical challenges will be critical for businesses to ensure the responsible use of AI in the workplace.

TERMS TO KNOW

Artificial intelligence (AI)
The branch of computer science that replicates human intelligence with minimal human interaction.

Robotics
The technology of machines that can work autonomously (without a human operator) to replicate the tasks of humans.